

MODERN THREAT HUNTING

10 PRACTICAL STEPS TO OUTSMART ADVERSARIES

A Hands-On Guide Using Hunt.io's Threat Intelligence Platform



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Executive Summary

This guide was built for threat hunters, SOC analysts, and security teams who need more than static IOCs and generic alerts. It outlines a practical, repeatable approach to discovering and mapping adversary infrastructure, built around real signals, behavioral patterns, and modern investigation techniques.

We walk through ten essential steps, from identifying a single malicious IP or certificate to expanding into broader clusters using ASN data, HTTP headers, SSH keys, and open directories. Each section includes live examples and HuntSQL™ queries to help you move from signal to insight without wasting time.

Whether you're pivoting on GoPhish headers, tracking reused SSH keys, or enriching IOC feeds with infrastructure intel, this guide is designed to sharpen your hunts and give you a tactical edge. No theory. Just methods that work.

Why This Guide Exists

Information systems, as we define them today, continue to be beset by a substantial number of both known and yet-to-be-discovered hardware and software vulnerabilities. This reality, however unsettling, is not hopeless. As we speak, scores of dedicated security professionals worldwide tirelessly navigate a complex web of security products and alerting instruments to bring clarity, containment, and resilience to environments under constant siege.

Until recently, traditional approaches relied on the reactive nature of many organizations, forcibly drawn to the notion of “bolt-on” security or to the overabundance of a class of breadcrumb intelligence hitherto known as *Indicators of Compromise* (IOCs). This guide exists to challenge such long-held practices.

Thus, we hope to elicit a lively discussion on the impact of threat hunting as the new epicenter of security operations, and to encourage you to make a deliberate shift away from any stitched solutions toward a solid, evidence-based preemptive program. In doing so, we will review some core concepts, highlighting Hunt’s role in supporting and accelerating our collective understanding of modern threats and the practical means by which we uncover them.

What is Modern Threat Hunting?

Modern threat hunting is a *proactive*, hypothesis-driven cybersecurity specialization conceived to uncover threats that may elude traditional detection mechanisms. This tenet operates on the assumption that adversaries -malicious entities attempting to compromise, disrupt, or gain unauthorized access to digital systems- already have a foothold in your environment and focuses on identifying potential traces of their activity.

A key task in this process is to encourage hunters to craft *investigative leads*: initial clues or signals that foster deeper inquiry into a set of specific behaviors across all available data sources. From endpoint telemetry and network flows to authentication and cloud logs, this idea hinges on the need to correlate disparate signals and map them to adversary tradecraft-namely, their tactics, techniques, and procedures (TTPs). These constitute a kind of behavioral “fingerprint” left by threat actors as they navigate an environment, escalate privileges, exfiltrate data, or attempt to evade detection, among other activities.

As hinted, modern threat hunting represents a **shift in mindset**. With organizations facing an increasing number of attacks, relying solely on baggy log correlation analysis or the good old static indicator belies the sophistication behind a significant portion of

them. Conversely, modern threat hunting emphasizes approaches such as *contextual intelligence and pattern recognition*-combined with the *analyst's intuition*, these cues behave as key catalysts in honing the detection life cycle.

While the latter are becoming more widely accepted, the platforms that enable this practice vary significantly in scope, depth, and purpose.

At Hunt.io, however, we believe threat hunting deserves more than a modest supporting role. In doing so, we offer several key differentiators tailored for modern hunters, including:

- **Proactive infrastructure discovery:** Hunt.io excels at identifying malicious activity early on, exposing potential threats through a series of high-fidelity IP and fingerprinting techniques.
- **IOC hunter:** Our platform automatically extracts and validates machine-readable IOCs from relevant research at scale, enabling faster and more reliable investigations, as well as the creation of actionable blocklists.
- **Support for multiple hunting approaches:** Hunt.io can handle structured, unstructured, and entity-driven hunts, making it adaptable to different analyst workflows and use cases.
- **Human-automation balance:** Hunt.io integrates analyst intuition with machine learning and advanced analytics, prioritizing fast signal processing while retaining human-led decision-making.
- **Robust web interface:** Our secure web interface delivers operational scale through bulk enrichment, advanced search, and real-time statistics.

These attributes position Hunt.io as a platform purpose-built for the demands of modern threat hunting, supporting both the drive and the discipline needed to stay ahead of advanced threats.

10 Practical Steps to Outsmart Adversaries

Building on these capabilities, Hunt.io provides analysts with a practical, no-nonsense, data-rich environment that enables them to uncover malicious infrastructure and monitor adversary activity, delivering depth and speed rarely matched by other platforms.

Whether you're pivoting off an IP address, dissecting TLS certificates, or correlating behaviors across ports and services, the Hunt toolset supports every stage of the modern hunting workflow.

But what sets Hunt.io apart isn't just the breadth of data; it's how the platform delivers, correlates, and transforms that very same data into action. For example, high-fidelity IP scanning, infrastructure fingerprinting, and domain-specific tools like HuntSQL™ enable analysts to move from speculation to insight in just a few steps.

By automating where it counts and preserving manual control where it matters, Hunt empowers analysts to operate with both accuracy and precision. The following ten steps outline a typical progression from a single lead to a broader, behaviorally anchored view of adversary infrastructure.

Let's begin.

Notes:

- *To run the examples in this guide, access to the Hunt.io platform is required.*
- *You can start a free 30-day trial here: <https://app-v1.hunt.io/register>*

Step 1 – Finding a Starting Point

Staring at an empty canvas is always intimidating, especially when that canvas spans worldwide infrastructure, intricate signals, and adversaries who are skillfully adept at covering their tracks. So, in such an increasingly cluttered scenario, how do we cut through the noise and find our way in?

Well, every hunt begins with an anchor, a concrete clue that cuts through noise and focuses your investigation. That might be:

- An IP address flagged in your threat feed
- A domain appearing in a recent compromise report
- An alert firing in your SIEM
- An IP making excessive connections to specific ports flagged by your firewall
- A suspicious file hash reported by your endpoint detection platform
- A certificate fingerprint observed in network traffic logs
- An anomalous login from an unexpected geolocation
- A phishing URL clicked by a user in your mail gateway

Public malware repositories often surface fresh indicators tied to adversary infrastructure, while real-time chatter on social media can reveal emerging C2 tactics before they land in traditional reports.

Once you've identified your anchor, treat it as a working hypothesis. Ask yourself:

- Where did this indicator show up?
- Which assets or protocols are involved?

Use those answers to drive your pivots -query passive DNS, review internal logs, or hunt TLS fingerprints- and turn that initial spark into a focused, hypothesis-driven investigation.

Step 2 – IOC Pivoting: Querying IPs in Hunt

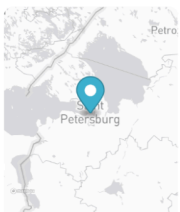
At the heart of infrastructure hunting lies a simple but powerful technique: IOC pivoting. By expanding from a single indicator, such as an IP address, analysts can trace connections, examine related assets, and begin to build a more exhaustive layout of adversary operations.

With an IP in hand, the next move is context. Hunt's IP view provides fast enrichment-geo, ASN, reverse DNS, and service history, all in one place. This context helps transform a one-off alert into a starting point for broader infrastructure mapping.

91.215.85.5 - Overview

Info
Domain 70
History
Associations 0
SSL History
SSH History
JARM
Port History
Signals Activity 0

91.215.85.5




PROSPERO OOO
Saint Petersburg, St.-Petersburg, RU

DNS

Reverse DNS -

Forward DNS -

Tag DNS -

ASN

IP Ranges 91.215.85.0/24

Hosting Companies PROSPERO OOO

ASN [AS200593](#)

Open Ports and Software

Name	Port	Vendor	Product	Version	Extra Info	First Seen	Last Seen	
SSH	22	-	-	-	-	2022-12-30	2025-04-10	Q
HTTP	80	Apache	HTTPD	2.4.58	-	2022-12-30	2025-04-09	Q
HTTP	443	Apache	HTTPD	2.4.58	-	2022-12-30	2025-04-15	Q

Begin with the **ASN**. Large consumer ISPs often point to compromised endpoints, while smaller ASNs-especially those tied to **bulletproof hosting**-can flag attacker-controlled infrastructure. Clicking the ASN in Hunt pivots you to other IPs in the same block for clustering.

Next, check **geolocation**. IPs resolving to unexpected regions-like Eastern Europe or Southeast Asia-may not be suspicious in isolation, but can raise flags when paired with malicious TLS certificates or non-standard ports.

Then review the **reverse DNS**. Hostnames resembling ISP pools (e.g., 123.45.67[.]89.ds1.example.net) usually involve residential traffic. In contrast,

names like `mail-update-control[.]net` may suggest deceptive infrastructure. Pivoting on the root domain or pattern surfaces related servers.

Each of these fields acts as a pivot point, letting you move laterally and build a clearer picture of the infrastructure behind the alert.

ASN & Domain Data Analysis

After reviewing an IP, examining its autonomous system number (**ASN**) helps uncover how it fits into a larger infrastructure footprint. ASNs provide insight into who owns the network space, their historical behavior, and whether the IP is part of a broader cluster of malicious activity.

91.215.85.5 - Overview

Info	Domain 70	History	Associations 0	SSL History	SSH History	JARM	Port History	Signals Activity 0
Hostname								Rank
 acc-update.info								-
 www.magical-burnell.91-215-85-5.plesk.page								-
 parcel-info.delivery								-
 suscripcion.info								-
 my.acc-update.info								-
 www.paquetes4-72co.info								-
 moncomptesnetflix.com								-
 www.express-upsdelivery.eu								-
 kpn.reactivering.info								-
 my.track-pakket.info								-
 ns1.meineabrechnung.info								-
 www.cranky-robinson.91-215-85-5.plesk.page								-
 track-view.info								-
 peaceful-chaum.91-215-85-5.plesk.page								-

This IP belongs to **Prospero OOO**, a **bulletproof hosting provider**, a term used to describe network operators that turn a blind eye to abuse complaints, allowing phishing sites, malware delivery, and other malicious activity to remain online.

Infrastructure hosted in these environments often plays a recurring role in cybercrime operations due to lax enforcement and rapid redeployment.

The domains tied to this server reinforce that pattern. Names like `parcel-info[.]delivery`, `acc-update[.]info`, and `moncomptesnetflix[.]com` suggest phishing campaigns designed to harvest credentials under the guise of tracking shipments or verifying accounts. Their co-location on a single IP within a permissive ASN suggests coordinated, reused infrastructure.

Hunt makes it easy to pivot across the ASN-surfacing other IPs, domains, and TLS certificates within the same network space. These pivots typically expose shared naming patterns, reused infrastructure templates, and links between operations that may initially appear unrelated.

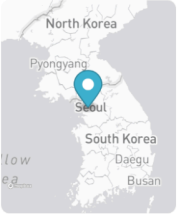
By layering in ASN-level context, defenders can identify hosting providers with a history of repeated involvement in malicious activity, helping distinguish isolated events from ongoing infrastructure abuse.

Investigating Ports and Services

While the previous section focused on infrastructure ownership and domain overlap, this step shifts to surface-level behavior, how a server presents itself on the internet and what that reveals about its role in an operation.

158.247.210.44 - Overview

Info
Domain 11
History
Associations 1
SSL History
SSH History
JARM
Port History
Signals Activity 0

158.247.210.44


The Constant Company, LLC
Paripark, Seoul, KR

DNS

Reverse DNS-

Forward DNS-

Tag DNS-

ASN

IP Ranges158.247.192.0/18

Hosting CompaniesThe Constant Company, LLC

ASN[AS20473](#)

Open Ports and Software

Name	Port	Vendor	Product	Version	Extra Info	First Seen	Last Seen	
SSH	22	-	-	-	-	2022-12-06	2025-03-21	Q
HTTP	80	Apache	HTTPD	2.4.52	-	2023-09-10	2025-04-13	Q
UNKNOWN	8081	-	-	-	-	2025-04-02	2025-04-02	Q

Open ports expose a system's functionality. Services like **SSH (22)** and **HTTP (80)** are expected, but they can still offer useful signals. In this example, the Apache HTTP server is publicly accessible, and its version number (**2.4.52**) may reveal unpatched vulnerabilities if not regularly maintained.

More interesting is port **8081**, labeled **UNKNOWN**; this may indicate a non-standard service or an intentionally misconfigured application, behaviors commonly seen in staging infrastructure. Threat actors often mask tools like admin panels, command-and-control payloads, or login portals behind high-numbered or uncommon ports.

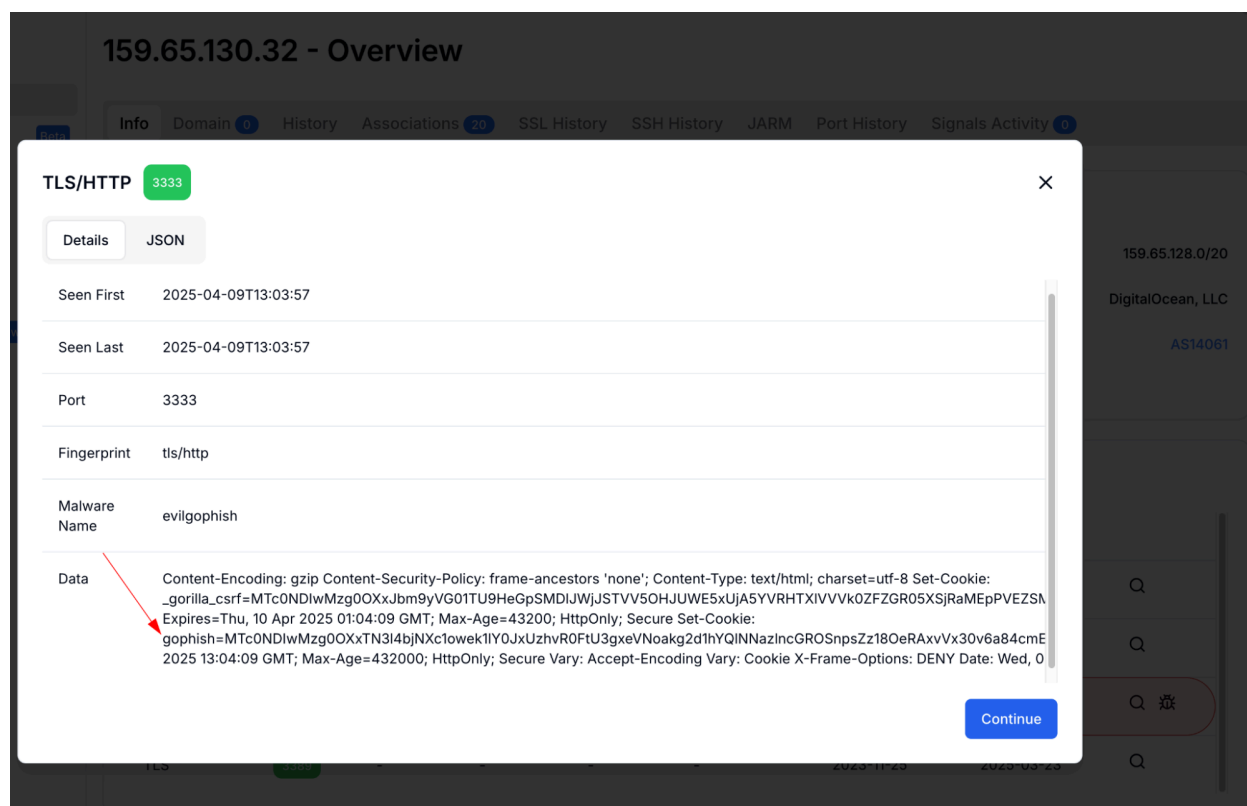
Even without interacting with the service directly, these passive observations can guide hunting priorities. A port with no banner, unexpected service behavior, or odd timing in historical data may merit closer inspection.

Step 3 – Fingerprinting Malware via HTTP Headers

HTTP headers can reveal consistent patterns in how malicious servers are configured and operated. Redirects, cookie values, and minimal responses often expose phishing kits, C2 frameworks, and redirectors.

The following examples show how these patterns can help identify and group related infrastructure.

Tracking GoPhish via structured HTTP response



In the above screenshot, the server on port 3333 presents a structured HTTP response tied to the phishing kit [GoPhish](#). While the response includes common fields like Content-Type and Content-Encoding, the key indicators are the Set-Cookie headers, specifically values like `__gorilla_csrf` and `gophish`, which are consistently present across deployments of this kit. These repeated values make it possible to track identical kits reused on unrelated infrastructure.

Let's now explore other practical examples.

Tracking Cobalt Strike Team Server via HTTP Headers

Requests to most Cobalt Strike team servers return nearly identical HTTP responses. These are intentionally minimal and designed to blend into the background noise of the internet; however, when viewed at scale, they are highly repeatable.

```
HTTP/1.1 404 Not Found
Content-Type: text/plain
Date: [Varies]
Content-Length: 0
```

This stripped-down response lacks banner identifiers or dynamic content, making it easy to overlook during casual inspection. However, its uniformity across deployments allows defenders to flag potential team servers that haven't been fronted by redirectors or traffic-shaping infrastructure.

Ports Protocols

Port	Protocol	Header Data
22	ssh	SSH-2.0-OpenSSH_7.4
80	http	<u>HTTP/1.1 404 Not Found Date: Fri, 28 Mar 2025 19:35:53 GMT Content-Type: text/plain Content-Length: 0</u>
4434	tls	
8082	http	HTTP/1.1 400 Bad Request Date: Mon, 3 Mar 2025 07:04:49 GMT Content-Type: text/plain Content-Length: 0
8088	http	HTTP/1.1 404 Not Found Date: Wed, 26 Mar 2025 21:03:41 GMT Content-Type: text/plain Content-Length: 0
8090	http	HTTP/1.1 400 Bad Request Date: Wed, 26 Feb 2025 19:34:51 GMT Content-Type: text/plain Content-Length: 0
8099	http	HTTP/1.1 400 Bad Request Date: Sun, 9 Mar 2025 17:05:38 GMT Content-Type: text/plain Content-Length: 0
8888	http	HTTP/1.1 400 Bad Request Server: nginx/1.18.0 Date: Fri, 04 Apr 2025 16:03:59 GMT Content-Type: text/html Content-Length: 157 Connection: close <html> <head><title>400 Bad Request</title></head> <body> <center> 400 Bad Request </center> <hr><center>nginx/1.18.0</center> </body> </html>

Identifying RedGuard Redirectors via HTTP Headers

Redirectors are often used to sit between a victim and the true command-and-control server, filtering traffic and masking backend infrastructure. One such example is [RedGuard](#), an open-source project designed to redirect or block suspicious HTTP

requests, typically used to protect Cobalt Strike team servers from unwanted scanning or direct access.

Ports Protocols		
Port	Protocol	Header Data
22	ssh	SSH-2.0-OpenSSH_9.2p1 Debian-2+deb12u5
80	http	HTTP/1.1 307 Temporary Redirect Content-Type: text/html; charset=utf-8 Location: https://office365.com Date: Mon, 13 Jan 2025 04:02:17 GMT Content-Length: 57 Temporary Redirect.
443	tls	HTTP/1.0 400 Bad Request Client sent an HTTP request to an HTTPS server.

The server returns an HTTP/1.1 307 Temporary Redirect to `https[:]//office365[.]com`-a tactic often used to mislead scanners by mimicking legitimate behavior. The Content-Type, Content-Length, and Location headers align with RedGuard's default configuration and, along with the consistent 307 status, can be used to fingerprint these redirectors. RedGuard servers also commonly reuse a default TLS certificate, which we explore further in Step 7.

Step 4 – Identifying Malicious Infrastructure by Certificate

TLS certificates are a [valuable asset](#) in infrastructure analysis. While domains and IP addresses often change to evade detection, certificates tend to linger, either due to operational reuse or deployment shortcuts. This persistence allows defenders to track connections that might otherwise remain hidden.

Whether issued by a public Certificate Authority or self-signed during setup, certificate metadata can reveal subtle infrastructure overlaps that support early detection and campaign tracking.

Certificate: 5951817150573092EE49AC05B53FF9BD6324268D80697AC15914821C5BB6670E

General
Details

Issued To

Common Name (CN)	Microsoft Windows Publisher
Organization (O)	Microsoft Corporation
Organization Unit (OU)	-

Issued By

Common Name (CN)	Microsoft Windows Publisher
Organization (O)	Microsoft Corporation
Organization Unit (OU)	-

Validity Period

Issued On	2024-11-29 03:46:10
Expires On	2025-11-29 03:46:10

Fingerprints

SHA-256 Fingerprint	5951817150573092EE49AC05B53FF9BD6324268D80697AC15914821C5BB6670E
SHA-1 Fingerprint	DF9A03B382351E59C73A93AD17E734CEDB536365

JA4X

JA4X	2bab15409345_2bab15409345_5318375af521
JA4XIssuer	2bab15409345
JA4XSubject	2bab15409345
JA4XExt	5318375af521

The certificate above claims to be for *.[microsoft.com](#), an attempt to impersonate Microsoft and create a façade of legitimacy around attacker-controlled infrastructure. It has been observed on infrastructure linked to **ShadowPad**, a modular backdoor commonly associated with Chinese state-aligned threat groups.

Certificate reuse is one of the more reliable techniques for linking seemingly unrelated servers. Many threat actors deploy the same certificate across multiple systems, often without modification. By pivoting on unique fields such as the subject common name, issuer details, or certificate serial number, analysts can uncover broader infrastructure supporting the same operation.

Even when certificates aren't reused exactly, certain anomalies can still provide valuable signals. For example:

- **Self-signed certificates** with generic subjects like CN=localhost often point to default tooling or hastily deployed servers.
- **Expired certificates** on active infrastructure may indicate neglected systems that are still used for malware staging or long-term C2.
- **Wildcard certificates** paired with typosquatting or brand-themed domains are commonly seen across phishing and malware delivery infrastructure.

The following examples show how defenders can use certificate metadata to track infrastructure more effectively.

Cobalt Strike Infrastructure by Certificate

Many Cobalt Strike team servers reuse self-signed or cloned TLS certificates with identifiable common names.

This [HuntSQL™](#) query looks for servers presenting certificates with a subject.common_name value of "**Major Cobalt Strike**", a known default string used in unmodified deployments of the framework.

```
SELECT ip, port, certificate_hash
FROM certificates
WHERE subject.common_name == 'Major Cobalt Strike'
GROUP BY ip, port, certificate_hash
```

This type of query helps defenders surface likely Cobalt Strike infrastructure, especially when paired with additional filters such as timeframe, ASN, or unusual ports. It's particularly useful for detecting poorly obfuscated team servers or lab environments exposed to the public internet.

Dashboard > SQL Explorer
Docs
Samples
MI

SQL Editor

```

1 SELECT
2   ip,
3   port,
4   certificate_hash
5 FROM
6   certificates
7 WHERE
8   subject.common_name == 'Major Cobalt Strike'
9 GROUP BY
10  ip,
11  port,
12  certificate_hash

```

Run Query

Format Query

Results
HTTP Malware Certificates Honeypot Open Directories Phishing Crawler Protocol URLx SSH Filenames JARM

Timeframe: timestamp.day gt '2025-04-13'

Download

63 Results

ip	port	certificate_hash
101.43.252.53	50050	9C7041032347CCA7335D614A75D1ABBB21ACAD11FD1EBF48DB986617CE5CF9A4
137.184.162.1	50050	F6FE73E1945C770E1C82075ADCA7D534B1ED3350BE658ACAC39EFD9FDF5B32FC
156.238.233.21	50050	AF493D219BF8DA8A86D27CBACAFB5D3BE8E5421D4A302F538BA4D59FC3D45218B
47.116.208.81	50050	56A06A233BD30F693DE25EF12CC19E8B2C92D3EB97DD969A2578DF084C376478
47.92.71.92	50050	3AAB7F87B445E9CECF16A6846E495F01AF387AD6BDD9F2F9C5C3E96B156A90ED
91.201.54.161	50050	5EDCBB9197963AF34D0BE7E95472004CA44FF31E6DDBC0354EF5ABE22B814FA3
60.204.144.169	50050	C80059CDD1D4D073B149BD812BF068650D259F42E1D1B172C646AA93028BF1E4

Each row represents an **IP:port** combination where a matching certificate was observed, giving analysts an actionable starting point to pivot into HTTP responses, reverse DNS records, or service behavior.

As you can see, when reviewed in context, TLS data becomes more than a passive artifact, it offers a durable signal for tracking infrastructure reuse and anticipating attacker behavior.

Step 5 - Finding Malware via SSH Keys

SSH key reuse remains one of the more persistent infrastructure artifacts. While domains and TLS certificates can change frequently, SSH fingerprints often persist across operations due to operator oversight or the use of automated deployment tooling; this makes them valuable identifiers for grouping infrastructure under a common operational umbrella.

Let's now explore some practical examples of SSH fingerprinting.

The fingerprint

4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e has been tied to **LockBit ransomware**, **RecordBreaker Stealer**, and **DanaBot**, and is currently observed across infrastructure spanning M247, Artnet, and other providers. Its repeated use highlights how SSH keys can persist across campaigns, making them useful for tracking threat actor infrastructure over time.

89.40.206.111 - Overview

Info	Domain 0	History	Associations 55	SSL History	SSH History	JARM	Port History	Signals Activity 0
------	----------	---------	-----------------	-------------	-------------	------	--------------	--------------------

Public SSH Keys 55	IOCs 0	Malware configs 0	Certificates 0
--------------------	--------	-------------------	----------------

Public SSH Keys					
IP ↕	Company ↕	Country ↕	SSH Fingerprint ↕	First Seen ↕	Last Seen ↕
37.28.156.23	Artnet Sp. z o.o.	PL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-18
37.120.237.200	M247 LTD Quebec Infrastructure	CA	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-20
38.132.108.251	M247 Europe SRL	US	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-16
45.84.121.38	M247 Europe SRL	NL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-22
78.135.73.154	COMNET BILGI ILETISIM TEKNOLOJILERI TIC. A.S.	TR	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-21
84.252.94.179	M247 LTD London Infrastructure	GB	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-21
89.40.206.90	M247 LTD New York Infrastructure	US	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-20
89.44.9.97	M247 Europe SRL	FR	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-17
89.238.185.20	M247 Ltd Amsterdam Infrastructure	NL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-22
91.206.178.69	Artnet Sp. z o.o.	PL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-04-09	2025-04-14
91.206.178.75	Artnet Sp. z o.o.	PL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-21
91.206.178.81	Artnet Sp. z o.o.	PL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-18
91.206.178.95	Artnet Sp. z o.o.	PL	4b8b3898c9034b14386799434c23ecfaba1fa96ba5051fcbd46344f380b15c2e	2025-03-24	2025-04-22

Let's now explore another example, this time tracking a specific threat actor.

Tracking Lazarous Group

The server at 91.222.173[.]30 has been linked to the **Lazarus Group**, a DPRK-affiliated threat actor, and is believed to be part of the infrastructure used in the [Bybit exchange breach](#). Within Hunt, the **Associations** tab reveals that the SSH key on this IP is reused across at least **88 other IP:port combinations**, many of which fall under the same provider, Virtual Systems LLC.

91.222.173.20 - Overview

Info	Domain 35	History	Associations 69	SSL History	SSH History	JARM	Port History	Signals Activity
Hostname								Rank
 cpcontacts.it.91-222-173-20.cprapid.com								-
 webdisk.it.91-222-173-20.cprapid.com								-
 autodiscover.coinbase-invoice.com								-
 autodiscover.dappradar.biz								-
 www.it.91-222-173-20.cprapid.com								-
 coinbase-invoice.com								-
 it.91-222-173-20.cprapid.com								-
 mail.91-222-173-20.cprapid.com								-
 www.91-222-173-20.cprapid.com								-
 webmail.it.91-222-173-20.cprapid.com								-
 mail.dappradar.biz								-
 webdisk.dappradar.biz								-
 mail.coinbase-invoice.com								-
 cpcontacts.coinbase-invoice.com								-

One of the IPs linked by SSH fingerprint -91.222.173[.]20- resolves to multiple phishing domains spoofing major cryptocurrency brands, including coinbase-invoice[.]com and dappradar[.]biz. These overlaps provide critical context to defenders tracking financially motivated operations or DPRK-attributed activity.

Unlike active scanning, which may be blocked or yield inconsistent results, passively collected SSH fingerprints offer a quiet and reliable way to uncover infrastructure relationships. They are particularly effective for identifying reused backend systems and mapping infrastructure that may not otherwise appear connected.

Step 6 - Tracking Malware Across a Known Abuse ASN

As discussed, ASNs provide a window into the ownership and operating history of network spaces, often revealing whether an IP belongs to a larger cluster of suspicious activity. So, when the same provider consistently appears in malware campaigns, it's rarely a coincidence.

For example, **M247 Europe SRL** is a popular hosting provider frequently leveraged by threat actors due to its global footprint and permissive abuse policies. By filtering Hunt's malware dataset by ASN, defenders can identify malware families repeatedly observed across this network.

SQL Editor

Run Query
Format Query

```

1 SELECT
2   ip,
3   hostname,
4   malware.name
5 FROM
6   malware
7 WHERE
8   asn.name == 'M247 Europe SRL'
9   AND timestamp > NOW - 30 DAY
10 GROUP BY
11   ip,
12   hostname,
13   malware.name

```

Results
HTTP Malware Certificates Honeypot Open Directories Phishing Crawler Protocol URLx SSH Filenames JARM

Custom Timeframe.

Download

199 Results

ip	hostname	malware.name
172.111.250.17	-	Remcos
213.199.38.61	-	Gophish
146.70.49.42	-	DcRat
213.199.38.61	www.cyber-intelligence.net	Gophish
217.156.50.222	-	Cobalt Strike
213.199.57.185	mail.queffroy.net	Gophish
37.120.151.102	-	Remcos
45.136.199.50	www.bb80.es	Keitaro

In this example, the query returns nearly 200 results from the past 30 days, revealing known malware such as **Remcos**, **Cobalt Strike**, **DcRat**, and **Gophish** deployed across various IPs and hostnames. These findings give researchers an immediate view into actor tooling and recurring payloads seen on the M247 infrastructure.

To generate these results, we can use the following HuntSQL™ query:

```
SELECT ip, hostname, malware.name
FROM malware
WHERE asn.name == 'M247 Europe SRL'
      AND timestamp > NOW - 30 DAY
GROUP BY ip, hostname, malware.name
```

This pattern can be adapted to other ASNs frequently abused in phishing, infostealer, or C2 operations, helping defenders monitor threat activity scoped to specific network providers.

Step 7 - C2 Infrastructure Tracking

Hunt's **C2 Feed** provides high-confidence visibility into active malicious infrastructure, delivered as a compressed JSON feed designed for automated ingestion.

The feed is built from internet-wide scans targeting malware protocols, suspicious TLS certificates, and fingerprints like **JARM** and **JA4**. Infrastructure on bulletproof networks is prioritized, and candidate servers are validated using custom logic to reduce false positives. Signatures are regularly updated by Hunt's research team.

Each feed entry includes structured threat intelligence, such as:

- **IP, port, hostname, and scan timestamp**
- **Malware family and subsystem** (e.g., Cobalt Strike)
- **Threat actor attribution**, including group name, aliases, country of origin, and targeting scope
- **IOC metadata**, such as SHA-256 hashes, publication sources, and associated analysis

The feed also surfaces infrastructure linked to reconnaissance, phishing, and other adversary staging activity. The data reflects the **last 7 days** of confirmed activity from the time the feed is requested.

To retrieve the feed via API:

```
curl -o c2.json.gz 'https://api.hunt.io/v1/feeds/c2' -H 'token:
<API_TOKEN_GOES_HERE>'
```

This output is ready for integration into internal alerting pipelines, enrichment engines, or daily hunting workflows with HuntSQL™.

The JSON output from the API query will look something like the below:

```

{
  "ip": "172.64.80.1",
  "port": 80,
  "hostname": "api.inmediavault.com",
  "timestamp": "2025-03-07T12:11:06",
  "scan_uri": "http://api.inmediavault.com/PEoX",
  "confidence": 100,
  "extra": {
    "status_code": 200,
    "geoip_city": null,
    "geoip_country": null,
    "geoip_asn": "CLOUDFLARENET",
    "geoip_asn_num": 13335,
    "geoip_subnet": "172.64.0.0/15",
    "domain_private_name": "inmediavault.com",
    "domain_type": "REGULAR"
  },
  "malware_name": "Cobalt Strike",
  "malware_subsystem": "C2",
  "ioc_hunter_metadata": [
    {
      "value": "d9e52663715902e9ec51a7dd2fea5241c9714976e9541c02df66d1a42a3a7d2a",
      "ioc_type": "ioc_sha256",
      "publication_name": "medium.com",
      "publication_title": "Shamoon Malware. Shamoon Malware is a wiper that spread...",
      "publication_url": "https://medium.com/@cyberecht/shamoon-malware-e24823501b10",
      "published_at": "2025-02-28T18:33:00",
      "threat_actor": {
        "id": 46,
        "uuid": "4f69ec6d-cb6b-42af-b8e2-920a2aa4be10",
        "name": "APT33",
        "origin": "APT33",
        "synonyms": "APT 33, Elfin, MAGNALLIUM, Refined Kitten, HOLMIUM, COBALT TRINITY, G0064, ATK35, Peach Sandstorm, TA451",
        "description": "Our analysis reveals that APT33 is a capable group that has carried out cyber espionage operations since at least 2013. We assess APT33 works at the behest of the Iranian government.",
        "meta": {
          "sponsor": "Iran (Islamic Republic of)",
          "victims": [
            "United States",
            "Saudi Arabia",
            "South Korea"
          ]
        }
      }
    }
  ]
}

```

IOC Hunter

PostHostsIPs



X

New XCSSET malware adds new obfuscation, persistence techniques to infect Xcode projects

Microsoft Threat Intelligence has discovered a new variant of XCSSET, a sophisticated macOS malware, which infects Xcode projects. This variant, first since 2022, features enhanced obfuscation, updated persistence mechanisms, and new infection strategies, making detection and removal more challenging. It steals files, system and user information, i... [Show More](#)

IOC Hosts:

bulknames.ru

IOC CVE:

CVE-2024-44243

Malware Files:

~/Library/Caches/com.apple.finder, /Applications/SimulatorTrampoline.app, /Applications/Reminders.app, /Applications/Reminders.app/Notes/, /Applications/Terminal.app, /Applications/Finder.app

Malware Name:

XCSSET

Create at:

2025-03-11T19:00:00

Update at:

2025-03-11T19:04:08

Created at

5 days ago

shark

Loader and

malware C2...

5 days ago

code projects.

gies, makin...

5 days ago

other routers

g, uses a...

5 days ago

ge malware

targeted use...

6 days ago

ge malware

e, exfiltrate...

6 days ago

Mozilla

malware is...

6 days ago

active IOCs -

6 days ago

The Strela Stealer Malware is being used by the SideWinder APT group to target Microsoft Outlook users for credential theft. SideWinder has expanded its scope to maritime and logistics companies, nuclear power plants, and energy infrastructure across Sou...

Stopping Sobolan Malware with Aqua Runtime Protection

Aqua Nautilus researchers have identified a new cyber attack campaign targeting Jupyter Notebooks and similar interactive computing

6 days ago

Querying the domain in Hunt revealed it resolved to 95.142.38[.]107, a server on the EuroByte LLC network in Russia. The IP was linked to over 50 domains, many of which aligned with indicators referenced in Microsoft's publication.

95.142.38.107 - Overview

Info
Domain 54
History
Associations 1
SSL History
SSH History
JARM
Port History
Signals Activity 0

Public SSH Keys 1
IOCs 0
Malware configs 0
Certificates 0

Public SSH Keys

IP	Company	Country	SSH Fingerprint	First Seen	Last Seen
95.142.38.79	MCHOST VPS Hosting	RU	241e5f70a46a3f0af5e62e8821b027fbd899e0e24a6e72917dc6b8db39b25138	2025-02-17	2025-03-09

However, Hunt's **Associations tab** revealed a key connection: a second IP, 95.142.38[.]79, which was not listed in the original reporting but shared the same public SSH key fingerprint. Pivoting on that fingerprint led to a third IP, 178.208.92[.]187, last observed in December 2024—likely part of the same infrastructure cluster supporting XCSSET operations.

Advanced Search

SSH.RSAHashSHA256:3FE88378D4EBDD3F27063D41DA0D4E69A0D3AAB549AEA28A0B8CD73DD9606891
Search

Filters
3 Results
Took: 1077ms

IP	Port	Hosting Company	HTTP Status	Last Seen	More details
95.142.38.107	22	EuroByte LLC		2025-03-17T04:29:00	
95.142.38.79	22	EuroByte LLC		2025-03-09T04:16:28	
178.208.92.187	22	EuroByte LLC		2024-12-10T05:09:13	

This example illustrates how a single domain from a threat feed can quickly expand into a broader view of the adversary's infrastructure by pivoting through metadata like SSH keys, hosting patterns, and shared certificates.

Step 8 - Tracking Phishing Pages

Phishing remains a significant attack vector across multiple verticals, consistently exploited by threat actors to gain initial access, harvest credentials, or distribute malware. And while email applications generally draw the most attention, the real infrastructure behind phishing campaigns -landing pages, redirect chains, and hosting patterns- offers defenders critical opportunities for detection.

In this step, we focus on detecting phishing infrastructure by identifying patterns commonly used in deceptive landing pages, like fake CAPTCHA prompts and malicious scripting.

Finding Click Fix-based Phishing Campaigns

[Click Fix](#) is a technique observed in phishing campaigns where users are prompted with fake CAPTCHA challenges-typically asking "Robot or Human?"-to initiate malicious behavior. Behind the scenes, these pages often trigger scripts using mshta to download or execute additional payloads.

The following HuntSQL™ query surfaces pages containing these known indicators:

```
SELECT timestamp, ip, url, body
FROM crawler
WHERE (body LIKE '%Robot or Human%' OR body LIKE '%mshta%')
GROUP BY timestamp, ip, url, body
```

The results can be used to track related infrastructure, identify recurring landing page structures, and cluster servers that reuse the same delivery mechanism.

Finding Phishing Pages by URL Pattern

Many phishing pages can be identified by simple keyword patterns in their URLs. This HuntSQL™ query searches the phishing dataset for URLs containing the word “login,” a common component in credential harvesting campaigns:

```
SELECT timestamp, url
FROM phishing
WHERE url LIKE '%-login%'
GROUP BY timestamp, url
```

SQL Editor

Run QueryFormat Query

```
1 SELECT
2   timestamp,
3   url
4 FROM
5   phishing
6 where
7   url like '%-login%'
8 GROUP BY
9   timestamp,
10  url
11
12
13
```

Results HTTP Malware Certificates Honeypot Open Directories Phishing Crawler Protocol URLx SSH Filenames JARM

Timeframe: timestamp.day gt '2025-03-22'

3,735 ResultsDownload

timestamp	url
2025-04-02T08:05:38	https://thaisafetywelding.com/olxtoto-login-link-alternatif-login-12/
2025-04-21T20:13:31	https://jt358-login.jtxportal.site/
2025-04-17T02:03:07	https://mail-login-refbbhadkchuhuyhcdyufgsgadfgcfthdvftgfdlbb.glitch.me/
2025-03-25T20:21:51	https://ah-ahsan-habib.github.io/mobimobile-Facebook-login-page/
2025-03-23T00:12:01	https://ngvmedia.net/altogel-login-17/
2025-04-21T20:33:45	https://megangregoryphotography.com/saham-toto-togel-login-deep-dive/
2025-03-28T08:55:35	https://reg-login.pages.dev/mawartoto
2025-04-17T13:12:25	https://kilau4d-alternatif-login.glitch.me/

The results return thousands of likely phishing pages, many of which impersonate login portals for services like email, banking, or web platforms. Queries like this are effective for surfacing newly registered or obscure domains used in phishing infrastructure.

Step 9 - Using Open Directories to Find Malware

Open directories -publicly accessible file repositories lacking proper access controls- are an attractive proposition for malicious actors seeking to host malware and similar artifacts under nominal scrutiny. To turn the tables on such a blind spot, defenders need tools capable of discovering and indexing exposed directories at scale.

[**AttackCapture™**](#) enables large-scale discovery of exposed files across open directories, helping defenders locate payloads, stolen data, and post-exploitation artifacts in real time.

In this step, we explore how open directories can reveal attacker behavior, especially when tools like Metasploit leave behind telltale folder structures that expose sensitive post-exploitation data.

Finding Metasploit Loot in AttackCapture™

One common pattern surfaced through AttackCapture™ is the `.msf4/loot/` directory, used by [**Metasploit**](#) to store data collected during exploitation. These directories often contain credential dumps, host enumeration logs, and keystroke captures.

Instead of scanning each directory manually, analysts can search for `.msf4/loot/` using the **FileName lookup** in Hunt's AttackCapture™ interface.

Search Result

File Name	▼	/.msf4/loot/	Search
Filename	File Size	Timestamp	
http://47.90.142.15:2333/.msf4/loot/20250104195808_default_200.135.4.71_host.windows.cle_914959.txt	0	Jan-09-2025 18:00:36	
http://209.38.167.196:8000/.msf4/loot/20241020215348_default_192.168.1.123_windows.system_579231.bin	18874368	Oct-21-2024 00:16:38	
http://209.38.167.196:8000/.msf4/loot/20241020215328_default_192.168.1.123_windows.sam_640990.bin	131072	Oct-21-2024 00:16:37	
http://209.38.167.196:8000/.msf4/loot/20241014222538_default_192.168.1.12_host.windows.key_207088.txt	717	Oct-15-2024 11:48:39	
http://111.229.0.18:8000/.msf4/loot/20231222124245_default_192.168.1.26_host.application_789228.txt	47543	Aug-09-2024 06:38:24	
http://111.229.0.18:8000/.msf4/loot/20240521104027_default_192.168.1.29_host.windows.cle_528212.txt	101	Aug-09-2024 06:25:44	
http://111.229.0.18:8000/.msf4/loot/20240521104137_default_192.168.1.29_host.windows.cle_958913.txt	0	Aug-09-2024 05:49:05	
http://103.170.122.93:443/.msf4/loot/20240626100325_default_10.102.23.52_linux.enum.users_683003.txt	4200	Jul-21-2024 03:19:27	
http://103.170.122.93:443/.msf4/loot/20240626100325_default_10.102.23.52_linux.enum.users_705481.txt	4328	Jul-21-2024 03:18:57	
http://103.170.122.93:443/.msf4/loot/20240626100258_default_10.102.23.52_linux.enum.users_614910.txt	9303	Jul-21-2024 03:18:06	
10 Rows per page		Page 1	

The results include base URLs, full file names, and timestamps, giving defenders a chance to investigate systems unintentionally leaking evidence of exploitation.

These findings are particularly useful in incident response and infrastructure monitoring, as they highlight not just what's hosted, but what may have already occurred.

Step 10 - Extending Hunts with the API

Hunt's expanded API suite enables teams to integrate live threat intelligence into SIEMs, enrichment pipelines, or automation workflows.

It supports flexible query parameters, real-time infrastructure lookups, and access to high-fidelity datasets including active C2s, phishing pages, and malware-hosting servers. With robust filtering and integration options, the API enables seamless threat detection at scale, turning Hunt's global visibility into actionable intelligence across your environment.

For example, this call retrieves active C2 infrastructure:

```
curl --request GET \  
      --url https://api.hunt.io/v1/c2s \  
      --header 'accept: application/json' \  
      --header 'token: <TOKEN>'
```

A snippet of the results:

```

{
  "active_c2s": [
    {
      "name": "Keitaro",
      "slug": "keitaro",
      "count": 5614,
      "count_str": "5,614",
      "categories": [
        "Management"
      ],
      "tags": [
        "Traffic Distribution System (TDS)",
        "Cross-platform"
      ],
      "short_description": "A Traffic distribution system (TDS) used by cybercriminals to redirect users to malicious sites, facilitating the spread of malware and phishing campaigns.",
      "long_description": "",
      "confidence_score": null,
      "confidence_tags": null,
      "icon_url":
"https://ui-avatars.com/api/?background=afe488&font-size=0.6&length=1&color=fff&name=Keita
ro"
    },
    {
      "name": "Tactical RMM",
      "slug": "tactical-rmm",
      "count": 3524,
      "count_str": "3,524",
      "categories": [
        "Management"
      ],
      "tags": [
        "Infostealer",
        "Windows"
      ],
      "short_description": "Tactical RMM is a remote monitoring & management tool, built with Django and Vue. It uses an agent written in golang and integrates with MeshCentral.",
      "long_description": "",
      "confidence_score": 95,
      "confidence_tags": "HTML match, title match",
      "icon_url":
"https://ui-avatars.com/api/?background=1d225d&font-size=0.6&length=1&color=fff&name=Tacti
cal RMM"
    }
  ],

```

HuntSQL™ queries can be adapted, shared, and reused across teams to build a consistent library of detection logic. Standardizing successful hunt patterns helps improve collaboration, streamline onboarding, and ensure valuable queries remain accessible and actionable over time.

Together, these practices help scale threat hunting from individual workflows to coordinated, intelligence-driven operations.

These ten steps provide a practical framework for modern threat hunting—from initial indicator pivots to infrastructure analysis and automation. Each step offers repeatable methods to uncover and track adversary activity at scale.

Summary

Modern threat hunting continues to undergo significant development under the banners of inquisitiveness, discipline, and adaptability. In other words, hunting is not a linear process, it is iterative, hypothesis-driven, and deeply rooted in behavioral context.

The ten steps outlined in this guide—from IOC pivoting and fingerprint analysis to ASN enrichment and TLS certificate correlation—offer a practical, repeatable strategy for uncovering adversary infrastructure at scale. Illustrating the pace at which adversaries continue to evolve and the complexities involved in tracking their whereabouts, Hunt empowers analysts to move beyond static detection and into a state of adaptive pursuit, where every pivot, anomaly, and shared artifact can unlock a larger narrative.

If you’ve made it this far, we encourage you to take the next step: explore Hunt.io yourself. Apply what you’ve learned here, test new ideas, see what works, and build your playbooks. In an era where adversaries move fast and hide well, only the curious, methodical, and adaptive hunters stay ahead.

Appendix

HuntSQL™ Cheat Sheet

A selection of high-value HuntSQL™ queries to accelerate your daily hunting and investigation workflows. Replace values where needed to fit your environment or focus.

Building Custom Queries

[HuntSQL™](#) allows defenders to interact directly with Hunt's underlying data, filtering, aggregating, and pivoting through massive volumes of security events using a simple, read-only SQL-like interface.

Designed for flexibility and speed, it allows analysts to tailor investigations to their threat models, whether tracking phishing pages, analyzing malware probes, or surfacing misconfigured services.

The screenshot displays the HuntSQL interface. At the top, there's a navigation bar with 'Dashboard > SQL Explorer', 'Docs', 'Samples', and a user profile 'MI'. Below this is the 'SQL Editor' section with a 'Run Query' button and 'Format Query' options. The SQL query entered is:

```
1 SELECT
2 *
3 FROM
4 malware
```

Below the editor is a tabbed interface with 'Results', 'HTTP', 'Malware', 'Certificates', 'Honeyplot', 'Open Directories', 'Phishing', 'Crawler', 'Protocol', 'URLx', 'SSH', 'Filenames', and 'JARM'. The 'Malware' tab is active, showing a table titled 'malware Command And Control (C2) servers tracked by Hunt.'.

Field	Description	Type	Example	Allowed Operators
description	Description	string	Show	◊ *
extra	Raw JSON data	string	Show	
tags	Tags	array	Show	
ip	IPv4 address	ip	Show	◊ <>
hostname	Hostname	string	Show	◊ *
botnet_name	Botnet name	string	Show	◊ *
endpoints	Endpoints	array	Show	
timestamp	Timestamp of the scan	date	Show	◊ <>
is_active	is active	bool	Show	◊

The interface supports standard SQL structure -SELECT, FROM, WHERE, and GROUP BY clauses- with useful aggregation functions like `uniq()`, `min()`, and `max()` to support temporal and behavioral analysis. Queries are scoped to datasets such as *http*, *protocol*, *malware*, and *certificates*, making it easy to target exactly the data needed.

For example, to review known malware events involving a specific IP:

```
SELECT *  
FROM malware  
WHERE ip == '192.168.1.1'
```

Malware Activity

Find known malware observed in the past 30 days:

```
SELECT ip, malware.name FROM malware WHERE timestamp > NOW - 30 DAY
```

List malware seen on frequently abused ASNs like M247 Europe SRL:

```
SELECT ip, malware.name FROM malware WHERE asn.name == 'M247 Europe  
SRL'
```

Open Directory Discovery

Find suspicious files by name (e.g., `passwords.txt`, `usernamest.txt`):

```
SELECT ip, malware.name FROM malware WHERE timestamp > NOW - 30 DAY
```

Identify exposed directories storing Cobalt Strike public/private key pairs for TeamServers :

```
SELECT * FROM open_directories WHERE file_name like  
'%cobaltstrike.beacon_keys%'
```

Certificate Tracking

List expired certs seen within the last 30 days:

```
SELECT ip, subject.common_name FROM certificates WHERE timestamp <
NOW AND timestamp > NOW - 30 DAY
```

Detect reuse of known, default certificates associated with Quasar RAT:

```
SELECT ip, port FROM certificates WHERE subject.common_name ==
'Quasar Server CA'
```

General Infrastructure Insights

Surface suspicious high-numbered HTTP ports:

```
SELECT ip, subject.common_name FROM certificates WHERE timestamp <
NOW AND timestamp > NOW - 30 DAY
```

Find servers hosting HTTP redirects:

```
SELECT * FROM http where (status.code = 301 or status.code = 302 or
status.code = 307)
```

Acknowledgements

Putting this guide together required more than good ideas; it required persistence, long hours, and a shared belief in the value of practical threat hunting.

Special thanks to Michael Rippey, Gianni Perez, Esteban Borges, Joe Dal Santo, Santiago Borges, and the rest of the Hunt.io team for shaping this work meaningfully. From structuring key concepts to refining technical examples, your input helped turn raw research into something practical and grounded.

We're also grateful to the broader threat hunting and intelligence community, the researchers who dig, document, challenge, and share. Your work helps all of us see further, think sharper, and stay ahead of what's coming.

The Hunt.io Team

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